

Treasury AI-Powered Alcohol Label Verification App

Project Design Document v2.0

Project Goal

Build a standalone web application that assists TTB compliance agents by automatically extracting information from alcohol label images, matching labels to application records, validating required fields, and generating explainable recommendations for human review.

The application is intended to reduce repetitive manual verification work while preserving human oversight and regulatory judgment.

The system functions as a decision-support tool and does not make final compliance determinations.

Success Criteria

A compliance agent can:

1. Upload application records through a CSV file.
 2. Upload one or more alcohol label images.
 3. Receive verification results within five seconds per label.
 4. Review suggested record matches.
 5. Identify mismatches quickly.
 6. Understand why a result was generated.
 7. Determine whether a label requires further review.
 8. Export results for additional analysis.
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Primary Workflow

Batch Verification Workflow

Upload Application CSV



Upload Label Images



OCR Extraction



Text Normalization



Record Matching



Field Validation



Recommendation Engine



Results Dashboard



CSV Export

Single Label Test Workflow

Single-label verification remains available for testing, demonstrations, and troubleshooting.

Upload Label



OCR Extraction



Field Validation



Results

This workflow is considered secondary to the batch workflow.

Required Fields

The prototype validates the common label elements shared across beverage categories.

Core Validation Fields

- Brand Name
- Class / Type
- Alcohol Content (ABV)
- Net Contents
- Government Warning Statement

These fields represent the primary validation scope of the prototype.

Supported Optional Fields

The following fields are supported for future extensibility:

- Bottler / Producer
- Country of Origin
- Beverage Type

These fields may be displayed or validated when data is available but are not primary compliance checks within the prototype.

Beverage Type Scope

TTB requirements vary by beverage category.

Supported values:

- Beer
- Wine
- Distilled Spirits
- Unknown

The prototype records and displays beverage type information but does not currently implement beverage-specific regulatory rule sets.

Future versions may implement:

Beer Validation Rules

Examples:

- Brewery-specific labeling requirements
- Alcohol content exceptions

Wine Validation Rules

Examples:

- Appellation validation
- Vintage validation
- Varietal validation

Distilled Spirits Validation Rules

Examples:

- Proof validation
- Age statement validation
- Category-specific requirements

This functionality is intentionally deferred to maintain focus on core verification workflows.

Validation Rules

Brand Name

Validation Type:

Fuzzy Match

Reason:

Minor capitalization and formatting differences should not automatically fail a label.

Example:

Expected:

STONE'S THROW

Detected:

Stone's Throw

Result:

Pass

Class / Type

Validation Type:

Fuzzy Match

Reason:

Formatting and capitalization differences are generally not compliance issues.

Alcohol Content

Validation Type:

Exact Match

Reason:

Alcohol content is a regulated value and must match application data.

Net Contents

Validation Type:

Exact Match

Reason:

Net contents must match application data.

Government Warning Statement

Validation Type:

Strict Match

Reason:

Federal regulations require specific warning language.

Record Matching Engine

Purpose

Identify the most likely application record for each uploaded label.

The matching engine compares OCR-extracted text against application records contained within the uploaded CSV.

Matching factors include:

- Brand Name
 - Class / Type
 - Alcohol Content
 - Net Contents
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Confidence Thresholds

Auto Match

95% and above

Action:

Automatically associate label with record.

Needs Review

70%–94%

Action:

Present suggested match to reviewer.

Human confirmation recommended.

No Match

Below 70%

Action:

Flag for manual review.

No automatic association is made.

Human Authority

Record matching recommendations are advisory only.

Final record assignment remains the responsibility of the compliance agent.

Recommendation Engine

The application may generate:

Pass

Field successfully matches expected data.

Fail

Field clearly does not match expected data.

Needs Review

Insufficient confidence exists to automatically determine a result.

Examples:

- Low-confidence OCR
- Ambiguous matching
- Missing information
- Poor image quality

The application intentionally favors human review when uncertainty exists.

User Types

Primary User

TTB Compliance Agent

Responsibilities:

- Upload records

- Upload labels
 - Review validation results
 - Make compliance determinations
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Secondary User

Compliance Supervisor

Responsibilities:

- Review escalated cases
 - Review exported reports
 - Monitor workflow effectiveness
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User Experience Goals

The application should be:

- Simple
- Fast
- Accessible
- Easy to learn
- Friendly to nontechnical users

Design principles include:

- Large, obvious controls
 - Minimal navigation
 - Progressive result display
 - Clear status indicators
 - Clear explanations
 - Minimal manual data entry
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Performance Goals

Individual Label Processing

Target:

Under 5 seconds

Batch Processing

Target:

Process 5–20 labels sequentially with visible progress feedback.

Results should appear as each label completes processing.

Application Startup

Target:

Under 3 seconds

Supported File Types

Label Images

Supported:

- PNG
 - JPG
 - JPEG
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Application Data

Supported:

- CSV

Required CSV fields:

- record_id
- brand_name
- class_type
- alcohol_content
- net_contents
- government_warning

Optional CSV fields:

- bottler_producer
 - country_of_origin
 - beverage_type
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Error Handling

OCR Failure

If OCR cannot confidently extract sufficient information:

Result:

Needs Review

Reason:

Poor image quality should not automatically fail a label.

Missing Required CSV Fields

The application displays a validation error and stops processing.

Example:

"The uploaded CSV is missing required fields: brand_name, alcohol_content."

No Match Found

Labels with no suitable application record are flagged for manual review.

Future Production Expansion

Potential future enhancements include:

- Beverage-specific validation engines
 - Font validation
 - Layout validation
 - Government warning placement analysis
 - Advanced image preprocessing
 - Audit logging
 - COLAs Online integration
 - Role-based access controls
 - Large-scale batch processing (200–500+ labels)
 - Secure document retention workflows
 - FedRAMP-aligned deployment architecture
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Out of Scope

The prototype intentionally excludes:

- COLAs Online integration
- Federal authentication
- Database persistence
- Case management
- Workflow approvals
- Enterprise deployment architecture
- External AI API dependency
- Final compliance decision automation
- Beverage-specific regulatory engines
- Font validation
- Label layout validation

These items are deferred to future development phases to maintain focus on the core verification workflow and prototype timeline.

Design Philosophy

This prototype prioritizes:

- Human-centered decision support
- Transparency
- Explainability
- Speed
- Simplicity
- Government-friendly architecture

The objective is not to automate regulatory judgment.

The objective is to reduce repetitive verification work so compliance agents can focus on higher-value review tasks.